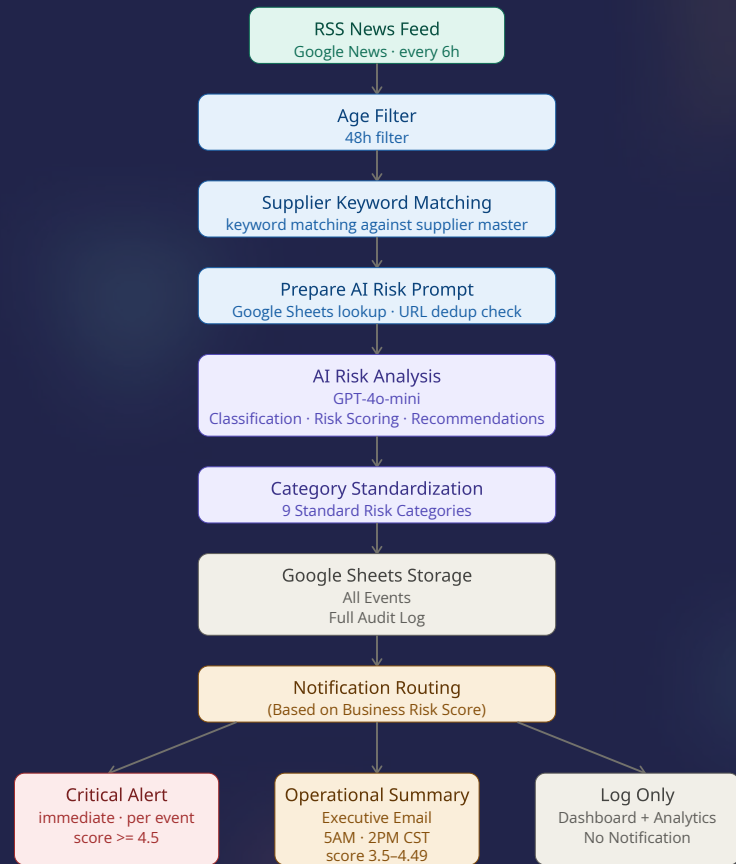




Gmail - source URL

Gmail - HTML table

Demo Scheduler: Every 6 Hours · Production Recommendation: Every 30 Minutes · Timezone: America/Chicago

AI Supply Chain Risk Intelligence Platform v2

AI-assisted workflow automation for supply chain risk monitoring, executive alerting, and operational intelligence – turning hours of manual effort into minutes of decisive action.

The Business Problem

Supply chain risk moves faster than email threads. By the time a critical supplier disruption surfaces through manual monitoring, the window for early response has often already closed.

Manual Monitoring Overload

Teams juggle scattered news sources, spreadsheets, and ad-hoc alerts with no consistent framework for triage or escalation.

Delayed Leadership Escalation

Critical events reach decision-makers too late — often after disruptions have already impacted production or logistics schedules.

Alert Fatigue & Noise

Without intelligent scoring, every event feels equally urgent — causing teams to tune out alerts entirely, missing the ones that matter most.

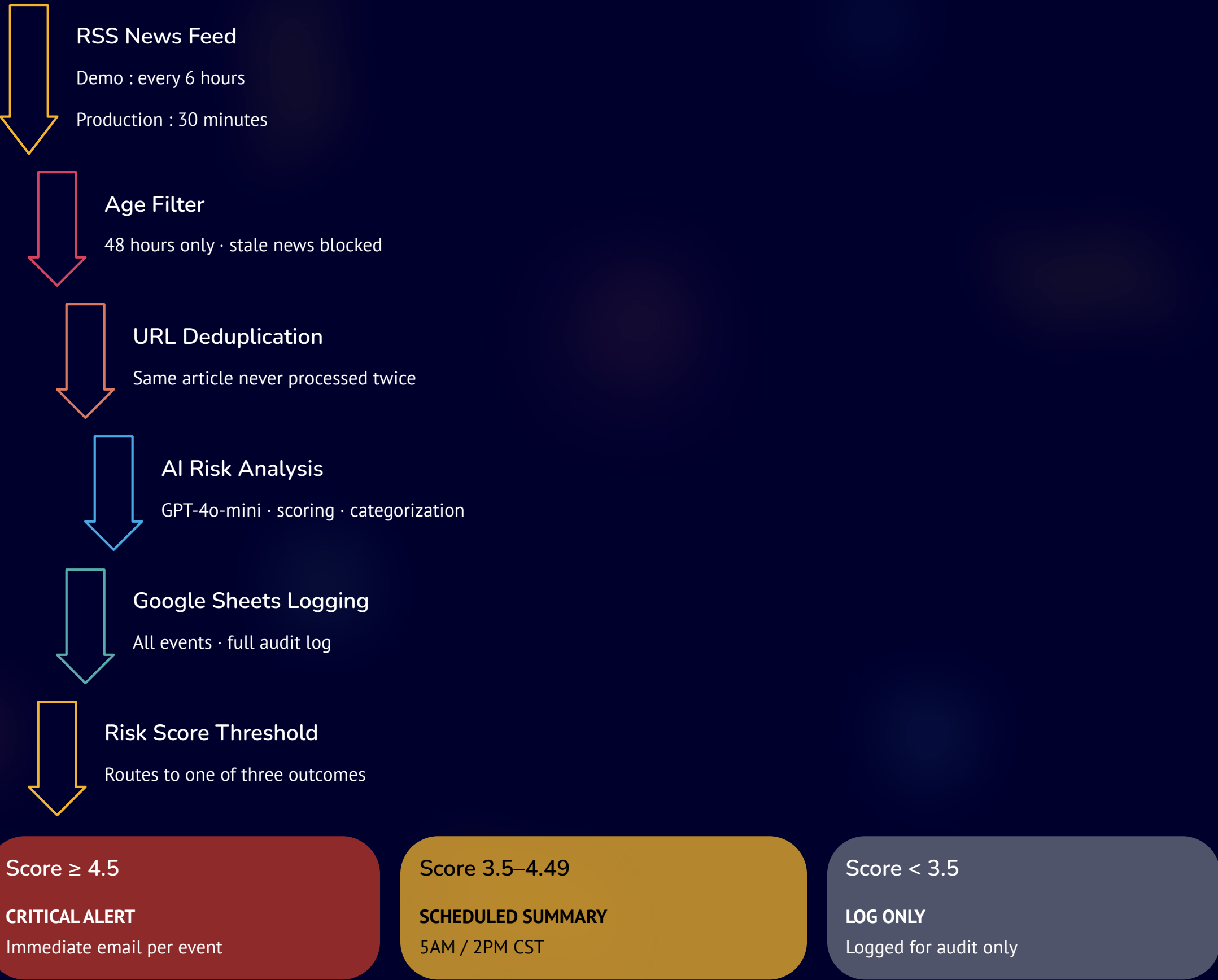
No Centralized Visibility

Risk data lives in inboxes, tabs, and silos. There is no single source of truth for supply chain risk status across the organization.

⊗ The cost of late response is measured in production delays, missed SLAs, and critical events that could have been avoided.

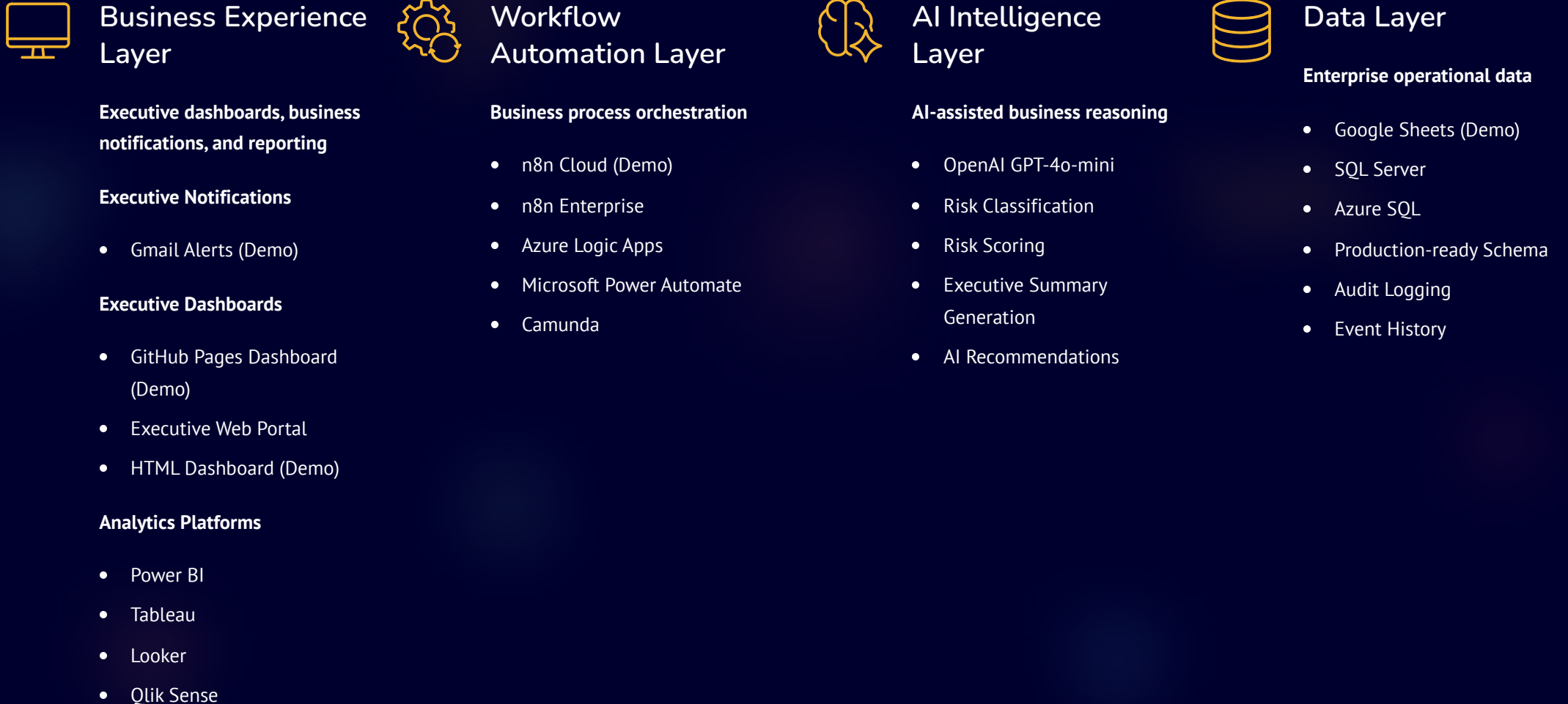
Solution Overview

From news to action in minutes, not days.



Enterprise Technology Architecture

Layered enterprise architecture enables scalable, maintainable, and technology-independent business solutions.



Layered Enterprise Technology Architecture decouples Business Experience, Workflow Automation, AI Intelligence, and Data into independently evolvable architectural layers, improving scalability, maintainability, governance, and long-term technology flexibility.



Critical Alert Workflow

High-risk events are escalated **immediately and automatically** – no human triage required to trigger an alert to leadership.

- 1 RSS Ingest**
Feeds polled every 30 min in production (every 6 hrs in demo)
- 2 Filter & Dedupe**
48-hour age filter applied; duplicate URLs suppressed before analysis
- 3 AI Scoring**
GPT-4o-mini evaluates severity, operational impact, and urgency
- 4 Immediate Alert**
Score ≥ 4.5 triggers instant email with source link and AI summary

✓ Every critical alert includes the source article as evidence – giving leadership the context to act, not just the notification to react.



Operational Summary Workflow

One email. Full operational picture. Twice a day. The summary workflow consolidates mid-tier risk events into structured digest emails designed for leadership consumption – not inbox overload.

 5:00 AM CST

Morning digest covering overnight events – optimized for Eastern time zone executives starting their day with situational awareness.

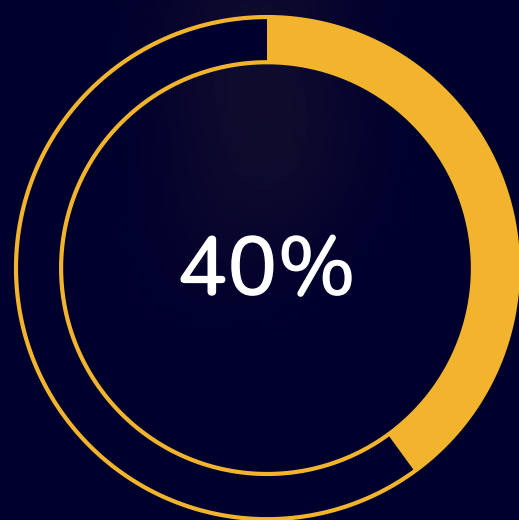
 2:00 PM CST

Afternoon digest for Pacific time zone coverage – ensuring no region misses a consolidated risk summary before end of business.

Each summary email contains an **AI-generated Executive Summary**, top-ranked risks with scores and supplier names, recommended actions, and a full source events table with clickable article links – everything a leader needs to make an informed call.

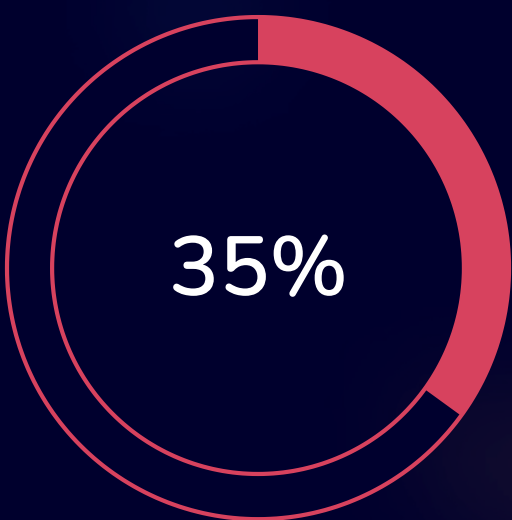
Risk Scoring Methodology

Scores reflect **business impact** — **not keyword frequency**. The weighted formula ensures that operational consequence drives prioritization, not just media volume.



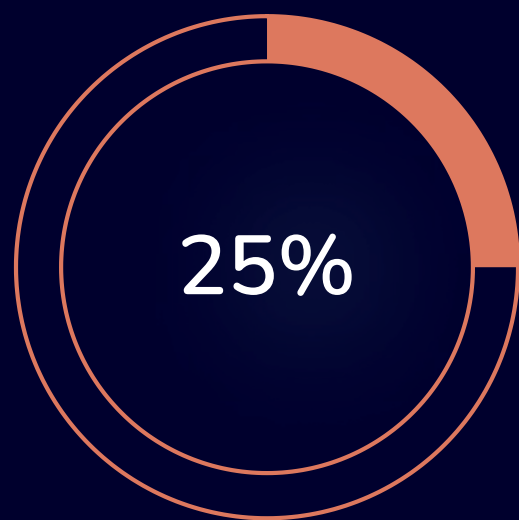
Severity

Magnitude of potential harm



Operational Impact

Disruption to production and logistics



Urgency

Response time requirement



Overall Score

Combined weighted result

 Score ≥ 4.5

CRITICAL — Immediate email alert sent per event. No delay, no batching.

 Score 3.5–4.49

SUMMARY — Included in next scheduled daily digest at 5AM or 2PM CST.

 Score < 3.5

LOG ONLY — Stored silently for audit and trend analysis. No notification generated.

Data Quality & Governance

Clean data. Auditable operations. Every event that enters the system is subject to quality controls before analysis – ensuring the AI never scores stale, duplicate, or irrelevant content.

1

Age Filter

Only articles published within the last 48 hours are processed. Stale news cannot trigger alerts or inflate risk scores.

2

URL Deduplication

Each article URL is hashed and checked before processing. The same story is never scored twice – preventing duplicate emails and inflated event counts.

3

Status Tracking

Every event records: `processed_at` · `alert_level` · `summary_status` · `critical_alert_status` · `dedupe_key`

A complete audit trail is maintained in Google Sheets – reviewable by any stakeholder without requiring access to the n8n workflow environment.

AI Capabilities

AI interprets risk. Humans decide. For every qualifying news event, the AI engine generates a structured, actionable intelligence package – not just a raw article link.



Risk Category

Classified into one of 9 standardized supply chain risk categories for consistent reporting.



Composite Score

Severity, Operational Impact, and Urgency scores generated independently, then combined via weighted formula.



Executive Summary

AI-drafted, leadership-ready narrative explaining the risk in plain business language – no translation required.



Recommended Action

Suggested next step for procurement or operations – flagged as AI-assisted, not autonomous.

 Confidence scores (0–100) accompany every AI output. Human review remains essential for all high-impact supply chain decisions.

Operational Data Layer

Every event logged. Every alert tracked. The data layer is designed for transparency today and enterprise migration tomorrow.

Why Google Sheets (Demo)

- Zero infrastructure setup – operational immediately
- Full transparency for stakeholder review
- Direct Power BI connectivity for dashboard integration
- SQL-ready schema – migration is designed and ready

23 Fields Tracked Per Event

Each logged event captures the full intelligence record including:

- Risk scores (Severity, Operational Impact, Urgency, Overall)
- Alert routing status and timestamps
- Deduplication key and source URL
- AI-generated summary, category, and recommended action
- Confidence score and processing metadata

Future migration to **SQL Server** is architecturally complete – the schema, field mapping, and query patterns are already designed for a seamless transition when production deployment begins.

Production Design Thinking

Designed for production. Deployed as a demo. Every architectural decision – from scheduling intervals to schema design – was made with enterprise-grade production requirements in mind, not demo convenience.

Component	Demo	Production
Scheduler	Every 6 hours	Every 30 minutes
Storage	Google Sheets	Enterprise Relational Database (SQL Server/Azure SQL)
Supplier Data	Fictional scenarios	ERP/Supplier Master Integration
Notifications	Gmail	Role-based Notifications (Email/Microsoft Teams/Slack)
Analytics	HTML prototype dashboard	Enterprise BI Platform (Power BI, Tableau, Looker, Qlik Sense)
Orchestration	n8n Cloud	Enterprise Workflow Platform (n8n Enterprise, Azure Logic Apps, Power Automate, Camunda)

✔ Core architecture, workflow design, and governance are production-ready.

Business Benefits

Operational value. Not just automation. The platform delivers measurable intelligence improvements across every layer of the supply chain risk function.

→ Reduced Manual Monitoring Effort

Automated ingestion and scoring replaces hours of daily manual news review across scattered sources.

→ Faster Critical Escalation

High-risk events reach leadership within minutes of publication – not hours or days after manual triage.

→ Reduced Alert Fatigue

Score-based routing ensures only genuinely critical events generate immediate notifications – quieting the noise without losing the signal.

→ Evidence-Based Decisions

Every alert includes source links, AI summaries, and confidence scores – giving leaders context, not just notifications.

PROCUREMENT

OPERATIONS

EXECUTIVE LEADERSHIP

SUPPLY CHAIN ANALYSTS



Future Roadmap

Each phase delivers independent business value while expanding automation, governance, and AI-driven intelligence.



Each phase delivers standalone business value while progressively increasing enterprise automation, governance, and operational intelligence.

About the Developer

Enterprise operations experience meets AI workflow automation. This platform was built by a practitioner who has spent 20+ years at the intersection of enterprise systems and operational process design.

Enterprise Systems

ERP, EDI, manufacturing, and logistics operations – implementation, integration, and day-to-day operational support.

Business & IT Bridge

Data exchange operations, cross-functional stakeholder alignment, and translating business requirements into technical solutions.

Operational Intelligence

Workflow automation, process improvement, and operational intelligence design – Now focused on Enterprise AI Automation Workflow Architecture Operational Intelligence Business Systems Analysis

Now focused on: **AI-powered workflow automation · Business process automation architecture · Enterprise operational intelligence**

Kaori Kashiwagi | GitHub: github.com/kk-2050



Design Philosophy

"The system separates event persistence, dashboard visibility, and human notification. All events are logged. Dashboards remain available for self-service monitoring. Only critical or summarized risks are pushed to users."

Separation of Concerns

Logging, alerting, and dashboarding are independent layers. Each serves a distinct purpose and audience – preventing coupling that creates fragility at scale.

Human-in-the-Loop

AI recommends. Humans decide. The platform amplifies human judgment – it does not replace it. Especially for high-impact supply chain actions.

Enterprise-Grade Reliability

Every design decision – from deduplication to audit tracking to schema design – is made for production-grade reliability, not demo convenience.

Designed for enterprise-grade reliability – not just automation.